

**E9 Trigonometric functions
in context A level only**

Name: _____

Class: _____

Date: _____

Time: **277 minutes**

Marks: **227 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

During a football match, a player kicks a ball from rest towards a goal, which is 12 metres from the ball. The ball is kicked with an initial velocity of 25 m s^{-1} at an angle of 17° above the horizontal.

- (a) Show that the time that it takes the ball to travel the horizontal distance of 12 metres is 0.502 seconds, correct to three significant figures.

(3)

- (b) The ball hits the crossbar of the goal. Calculate the height of the crossbar.

(3)

- (c) Find the angle between the velocity of the ball and the **vertical** just before the ball hits the crossbar.

(5)

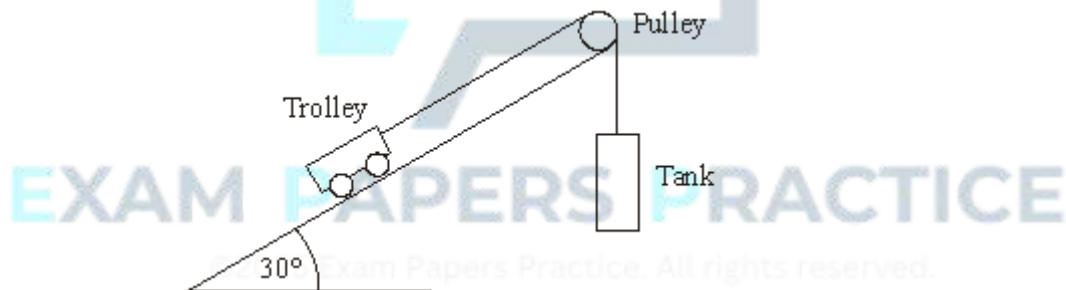
- (d) State two modelling assumptions that you have made.

(2)

(Total 13 marks)

Q2.

A trolley, of mass 200 kg, is on a slope inclined at 30° to the horizontal. It is attached to a tank by a light, inextensible rope that passes over a smooth, light pulley. The tank can be filled with water. The situation is shown in the diagram below.



Model the tank and trolley as particles and assume that there is no resistance to motion.

- (a) When the tank is filled with water, the total mass of the tank and the water is 150 kg.

Show that the acceleration of the tank and trolley is 1.4 m s^{-2} .

(7)

- (b) When the tank is empty, the trolley can be set in motion, so that it moves with a constant speed up the slope. Find the mass of the empty tank.

(4)

(Total 11 marks)

Q3.

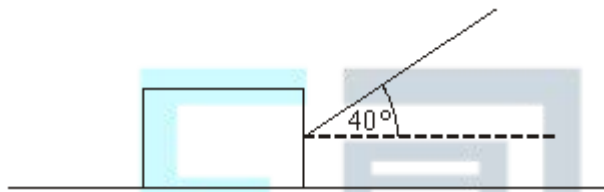
A ball is kicked from ground level so that its initial velocity is 10 m s^{-1} , at an angle of 60°

above the horizontal. It hits a wall, which is at a distance of 8 metres from the initial position of the ball.

- (a) Show that the ball hits the wall 1.6 seconds after it was kicked. (3)
 - (b) Find the height of the ball as it hits the wall. (3)
 - (c) Find the speed of the ball as it hits the wall. (5)
- (Total 11 marks)**

Q4.

A box, of mass 50 kg, is dragged along rough horizontal ground by a rope attached to the box. The rope makes an angle of 40° with the ground and the tension in the rope is T newtons. The coefficient of friction between the box and the ground is 0.6. The diagram shows the box and the rope.

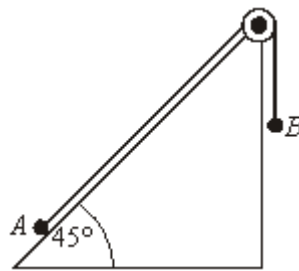


Model the box as a particle.

- (a) Draw a diagram to show the forces acting on the box. (1)
 - (b) Find, in terms of T , the magnitude of the normal reaction force acting on the box. (3)
 - (c) Show that the magnitude of the friction force acting on the box is $294 - 0.6T \sin 40^\circ$. (2)
 - (d) The box accelerates at 0.5 m s^{-2} . Find T . (4)
- (Total 10 marks)**

Q5.

Two particles, A and B , are connected by a light inextensible string, which passes over a smooth light pulley. Particle A is on a smooth slope, at 45° to the horizontal, and particle B hangs with the string vertical, as shown in the diagram.



The mass of A is 14 kg and the mass of B is 6 kg.

- (a) Using two equations of motion, show that the acceleration of the particles is 1.91 m s^{-2} correct to three significant figures. (6)
- (b) Particle B is replaced by a particle C of mass m kg. After the particles have been set in motion, they move with a constant speed. Find m . (4)

(Total 10 marks)

Q6.

A crate, of mass 50 kg, is at rest on a warehouse floor. The floor is rough and horizontal. The coefficient of friction between the crate and the floor is μ . A rope is attached to the crate at an angle of 30° to the horizontal. The tension in the rope is 100 N. The crate is shown in the diagram.



Model the crate as a particle.

- (a) Draw and label a diagram to show the forces acting on the crate. (1)
- (b) Show that the magnitude of the normal reaction force acting on the crate is 440 N. (3)
- (c) If the crate remains at rest, μ must satisfy the inequality $\mu \geq k$. Find k . (3)
- (d) If $\mu = 0.1$, find the acceleration of the crate. (4)

(Total 11 marks)

Q7.

A ball is thrown so that it initially travels at 10 m s^{-1} at an angle of 70° above the horizontal.

- (a) A simple model assumes that the ball is thrown from ground level and lands at the same level. Find the time of flight and the range of the ball.

(5)

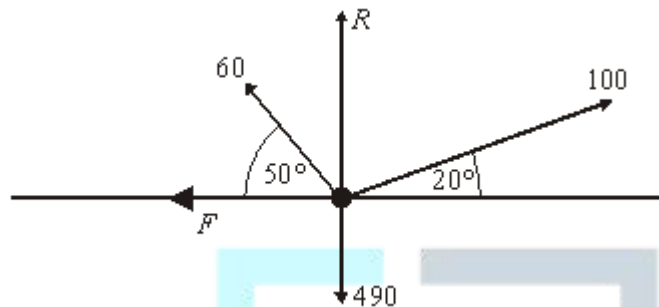
- (b) A refined model assumes that the ball is at a height of 2 metres when it is thrown, and lands at ground level. Find the range of the ball based on this assumption.

(7)

(Total 12 marks)

Q8.

A box, of mass 50 kg, moves in a straight line on a rough horizontal surface. The diagram below shows **all** the forces acting on the box, as it moves. The magnitude of each force is in newtons. The box is modelled as a particle.



Note that the weight of the box has been included in the diagram.

- (a) Find R , the magnitude of the normal reaction force acting on the box.

(3)

- (b) The box accelerates at 0.5 m s^{-2} .

- (i) Show that F , the magnitude of the friction force acting on the box, is approximately 30.4.

(4)

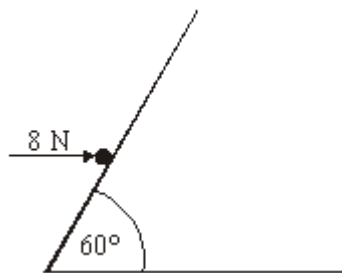
- (ii) Find the coefficient of friction between the box and the surface.

(2)

(Total 9 marks)

Q9.

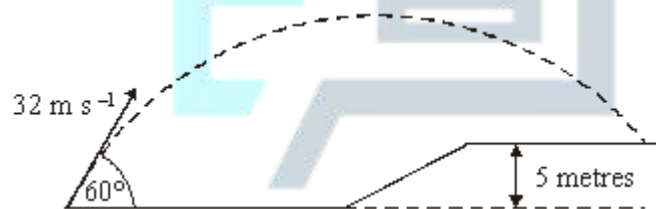
A particle is held at rest on a smooth slope by a horizontal force of magnitude 8 newtons, as shown in the diagram below. The slope is at an angle of 60° to the horizontal.



- (a) Draw a diagram to show the forces acting on the particle. (1)
 - (b) By resolving horizontally show that the magnitude of the normal reaction force acting on the particle is approximately 9.24 newtons. (3)
 - (c) Find the mass of the particle, giving your answer to two significant figures. (3)
- (Total 7 marks)**

Q10.

A golf ball is hit so that it initially travels at 32 m s^{-1} at an angle of 60° above the horizontal. The ball lands on a horizontal surface 5 metres higher than the ground from which it was hit, as shown in the diagram below.



- (a) Show that the ball is in the air for approximately 5.47 seconds. (5)
 - (b) Calculate the horizontal distance travelled by the ball. (2)
 - (c) Find the speed of the ball when it hits the ground. (5)
- (Total 12 marks)**

Q11.

A particle of mass 60 kg is on a rough surface inclined at an angle of 40° to the horizontal.

- (a) Find the magnitude of the normal reaction force acting on the particle. (2)
- (b) If the particle remains at rest, find the minimum value of the coefficient of friction between the particle and the slope.

(4)

- (c) If the coefficient of friction between the particle and the slope is 0.2, the particle slides down the slope. Find the acceleration of the particle in this case.

(5)

(Total 11 marks)

Q12.

A football is kicked from horizontal ground. It initially moves with speed 20 m s^{-1} at an angle of 30° above the horizontal.

- (a) (i) Show that the ball hits the ground approximately 2.04 seconds after it has been kicked.

(4)

- (ii) Hence find the range of the ball.

(2)

- (b) In fact the ball comes into contact with a player's head when it is at a height of 2 metres. Find the speed of the ball at this height.

(8)

(Total 14 marks)

Q13.

A block, of mass 5 kg, is held at rest on a rough plane, which is inclined at 30° to the horizontal. The block is released and slides down the plane. The coefficient of friction between the block and the plane is 0.2.

- (a) Draw a diagram to show the forces acting on the block as it slides.

(1)

- (b) Show that the magnitude of the friction force acting on the block is approximately 8.5 N.

(3)

- (c) Find the acceleration of the block.

(4)

- (d) Find the speed of the block when it has travelled 1.2 metres down the slope.

(3)

(Total 11 marks)

Q14.

A particle is projected from a horizontal surface at a speed V and at an angle α above the horizontal.

- (a) Prove that the maximum height of the particle is $\frac{V^2 \sin^2 \alpha}{2g}$.

(6)

- (b) A ball is hit from ground level. The ball initially moves at an angle of 60° above the horizontal. The maximum height of the ball is 6 metres above the ground. Modelling the ball as a particle:

(i) find the initial speed of the ball;

(3)

(ii) find the range of the ball.

(4)

(Total 13 marks)

Q15.

A van has mass 1200 kg. It travels with constant acceleration up a slope inclined at 4° to the horizontal. The length of the slope is 250 metres. At the bottom of the slope the van has a speed of 20 m s^{-1} and at the top its speed has dropped to 15 m s^{-1} .

- (a) Show that the acceleration of the van while it is on the slope is -0.35 m s^{-1}

(3)

- (b) A simple model assumes that no external resistance forces act on the van. A constant force of magnitude P newtons parallel to the slope acts on the van.

Find P .

(4)

- (c) A more realistic model assumes that a resistance force does act on the van. This force has magnitude 300 newtons and acts parallel to the slope. Revise your answer to part (b) to take account of this extra force.

(2)

- (d) In reality the resistance force will not be constant. Explain why.

(2)

(Total 11 marks)

Q16.

A skier slides in a straight line directly down a slope inclined at 30° to the horizontal. The coefficient of friction between her skis and the slope is 0.3. The skier and her equipment are to be modelled as a particle of mass 80 kg. Assume that there is no air resistance present.

- (a) Draw a diagram to show the forces acting on the skier.

(1)

- (b) (i) Find the magnitude of the normal reaction force acting on the skier.

(2)

- (ii) Show that the magnitude of the friction force acting on the skier is 204 N to three significant figures.

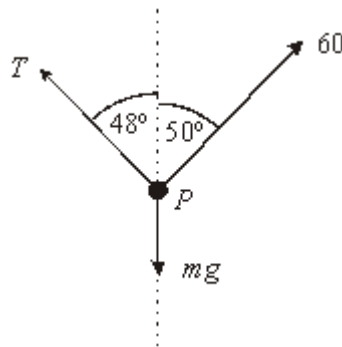
(2)

- (c) Find the acceleration of the skier.

(4)
(Total 9 marks)

Q17.

A particle, P , of mass m kg, is held in equilibrium by two strings. One string is inclined at 50° to the vertical and exerts a force of 60 newtons on the particle. The other string exerts a force of magnitude T newtons at an angle of 48° to the vertical. The forces that act on the particle are shown in the diagram below.



- (a) Find T .

(3)

- (b) Find m .

(4)

(Total 7 marks)

Q18.

A particle is projected from a horizontal surface and lands on the same horizontal surface. Its initial velocity is V m s⁻¹ at an angle α above the horizontal.

- (a) Prove that the range of the particle is

$$\frac{V^2 \sin 2\alpha}{g}$$

(6)

- (b) Hence explain why the maximum range of the particle is $\frac{V^2}{g}$, and state the value of α for which the particle has this maximum range.

(2)

- (c) A machine for launching tennis balls fires the balls from ground level on a horizontal surface. Assume that the balls are all fired at the same speed. The maximum range of one of these balls is 35 metres.

- (i) Find the speed at which the tennis balls are fired.

(2)

- (ii) Find the range of a tennis ball fired at an angle of 30° above the horizontal, giving your answer to the nearest metre.

(2)

(Total 12 marks)

Q19.

A rope is used to pull a box, of mass 20 kg, up a rough slope. Assume that the box moves with a constant acceleration.

- (a) As the box moves 4 metres up the slope its speed increases from 1 m s^{-1} to 5 m s^{-1} . Calculate the acceleration of the box.

(3)

- (b) The slope is at an angle of 30° to the horizontal. Assume that the rope is parallel to the slope. The coefficient of friction between the box and the slope is 0.4.

- (i) Draw a diagram to show the forces on the box.

(1)

- (ii) Show that the magnitude of the friction force acting on the box is approximately 68 N.

(3)

- (iii) Find the tension in the rope.

(4)

(Total 11 marks)

Q20.

A block, of mass 7 kg, is placed on a rough slope that is inclined at 30° to the horizontal, as shown in the diagram. The block remains at rest in this position.



- (a) Draw a diagram to show the forces acting on the block.

(1)

- (b) Find the magnitude of the normal reaction force acting on the block.

(2)

- (c) Find the magnitude of the friction force acting on the block.

(2)

- (d) The coefficient of friction between the block and the plane is μ . Find an inequality that μ must satisfy.

(2)

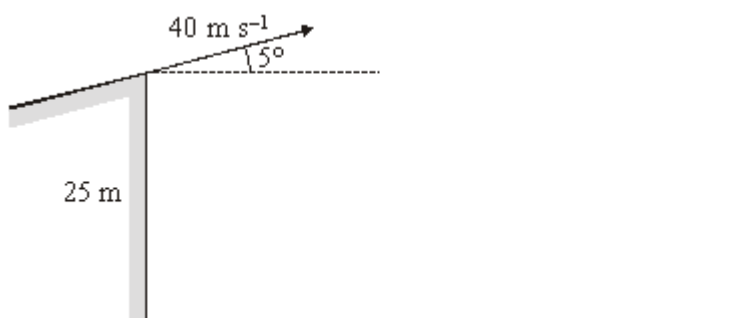
- (e) A similar block, of mass 14 kg, is placed on the slope. Does this block remain at rest or slide? Give a reason for your answer.

(2)

(Total 9 marks)

Q21.

In a film a stunt-man drives a car off the top of a vertical cliff. The top of the cliff is 25 metres above the level of the sea. When it leaves the cliff the car is travelling at 40 m s^{-1} at an angle of 5° above the horizontal. The diagram shows the cliff and the initial velocity of the car.



Model the car as a particle and assume that, while it is in the air, it moves under the influence of gravity alone.

- (a) Show that the car hits the sea approximately 2.64 seconds after it leaves the top of the cliff.
- (b) Find the horizontal distance of the car from the cliff when the car hits the sea.
- (c) Find the speed of the car when it hits the sea.

(6)

(2)

(5)

(Total 13 marks)